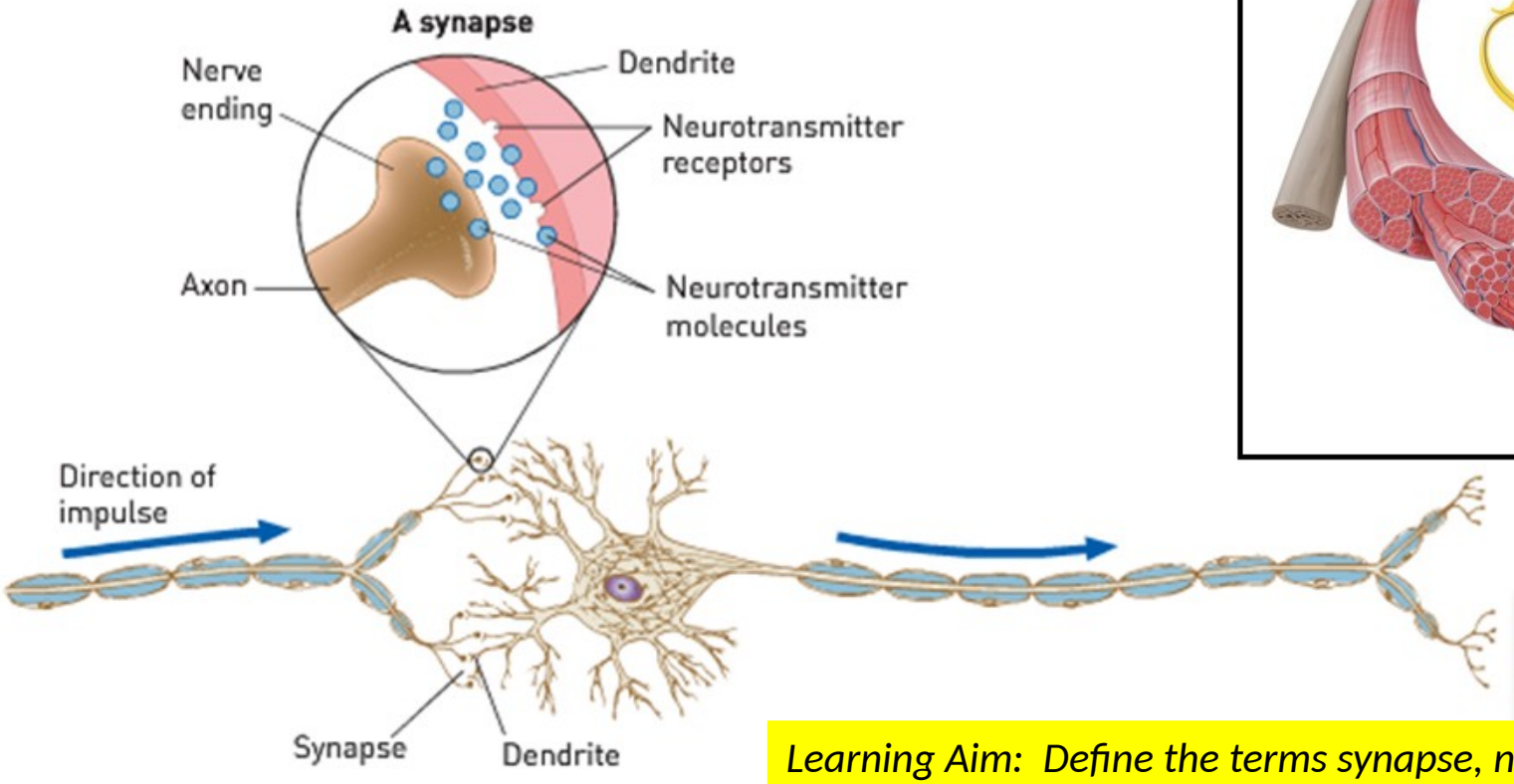
Synapses and Neurotransmitters

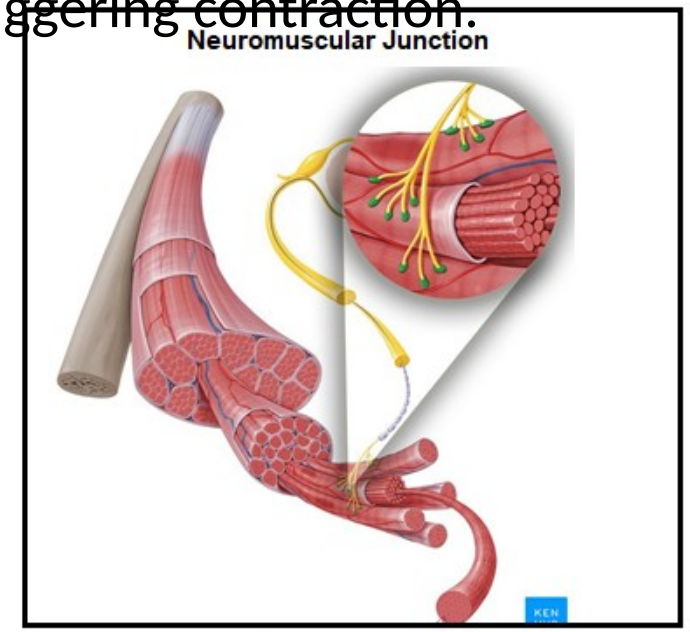


Nerve Cells and Nerve Impulses

Synapses and Neurotransmitters Overview

- ***Synapse***: a small gap between one neuron and the next.
- Messages passed across the gap by release of chemicals called ***neurotransmitters***.
- ***Neuromuscular junction*:** where an axon terminal meets muscle tissue instead of another neuron – impulses passed from nerve to muscle, triggering contraction.

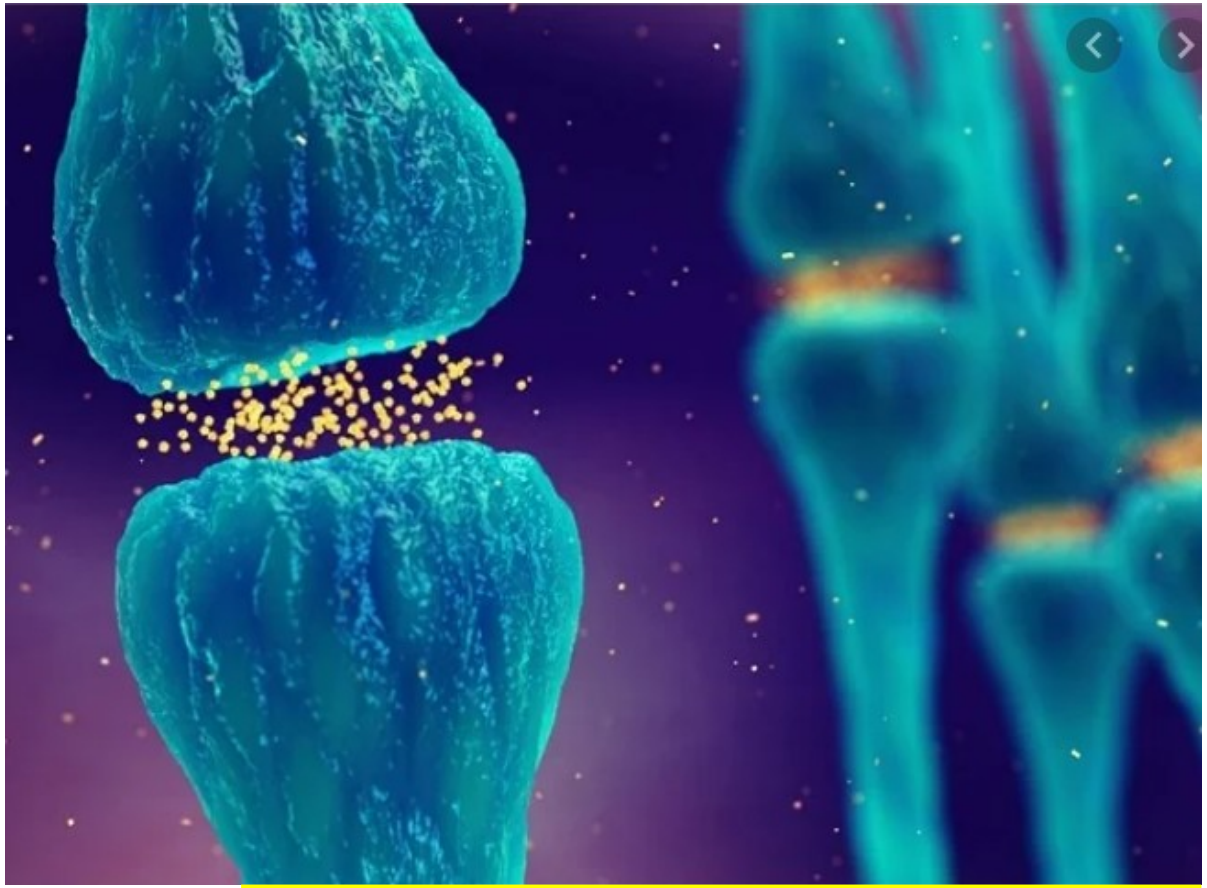




Learning Aim: Define the terms synapse, neurotransmitter and neuromuscular junction.

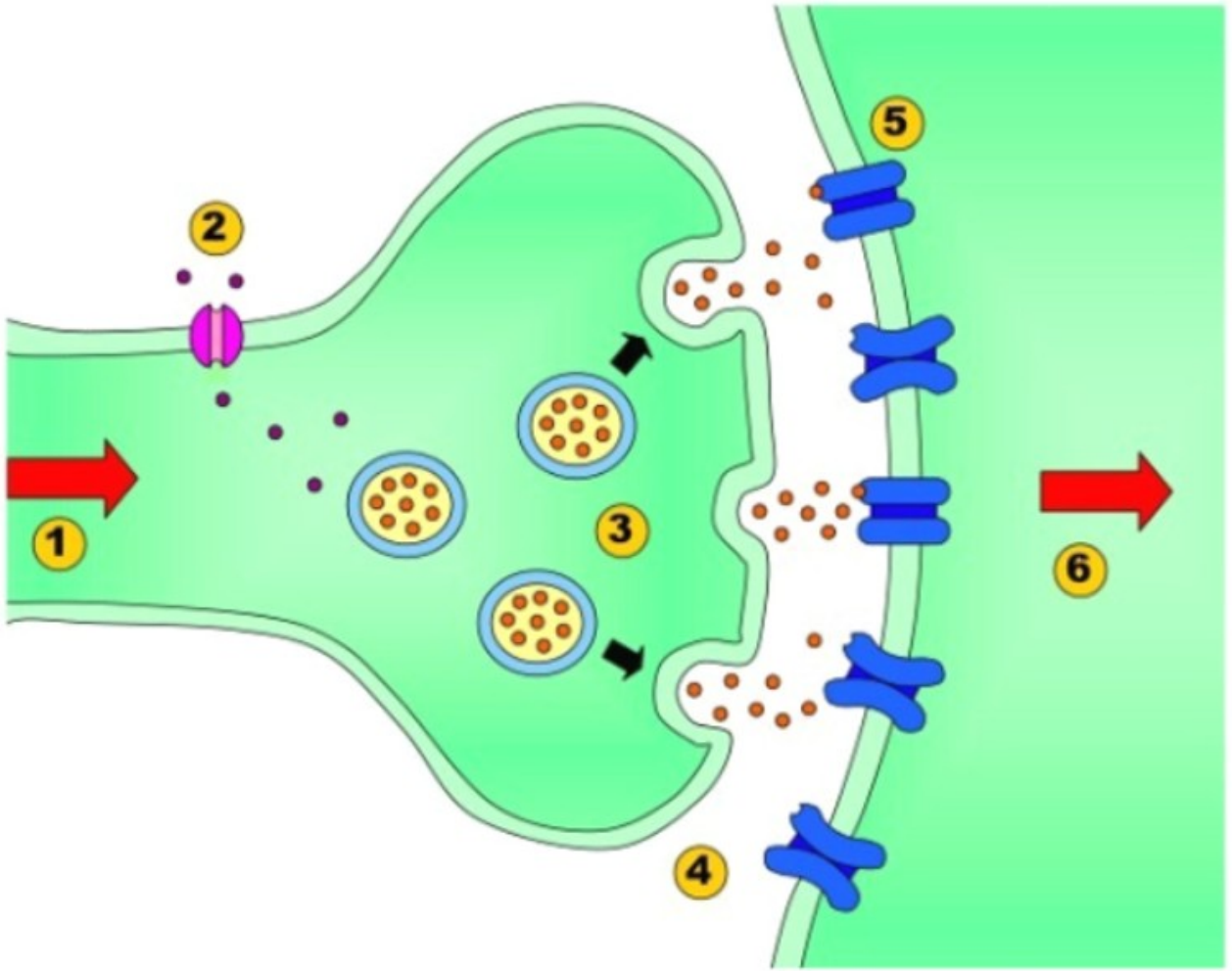
Neurotransmitters

- Released from the axon terminals in response to action potential
- Diffuse across the synapse
- Attach to receptors on the dendrite of next neuron
- Include chemicals such as:
  - Acetylcholine (ACh)
  - Noradrenalin (Norepinephrine)
  - Adrenaline (Epinephrine)
  - Dopamine
  - Histamine
  - Adenosine
  - Serotonin



Learning Aim: List some common neurotransmitters.

Transmission of an Action Potential (AP) across the synapse

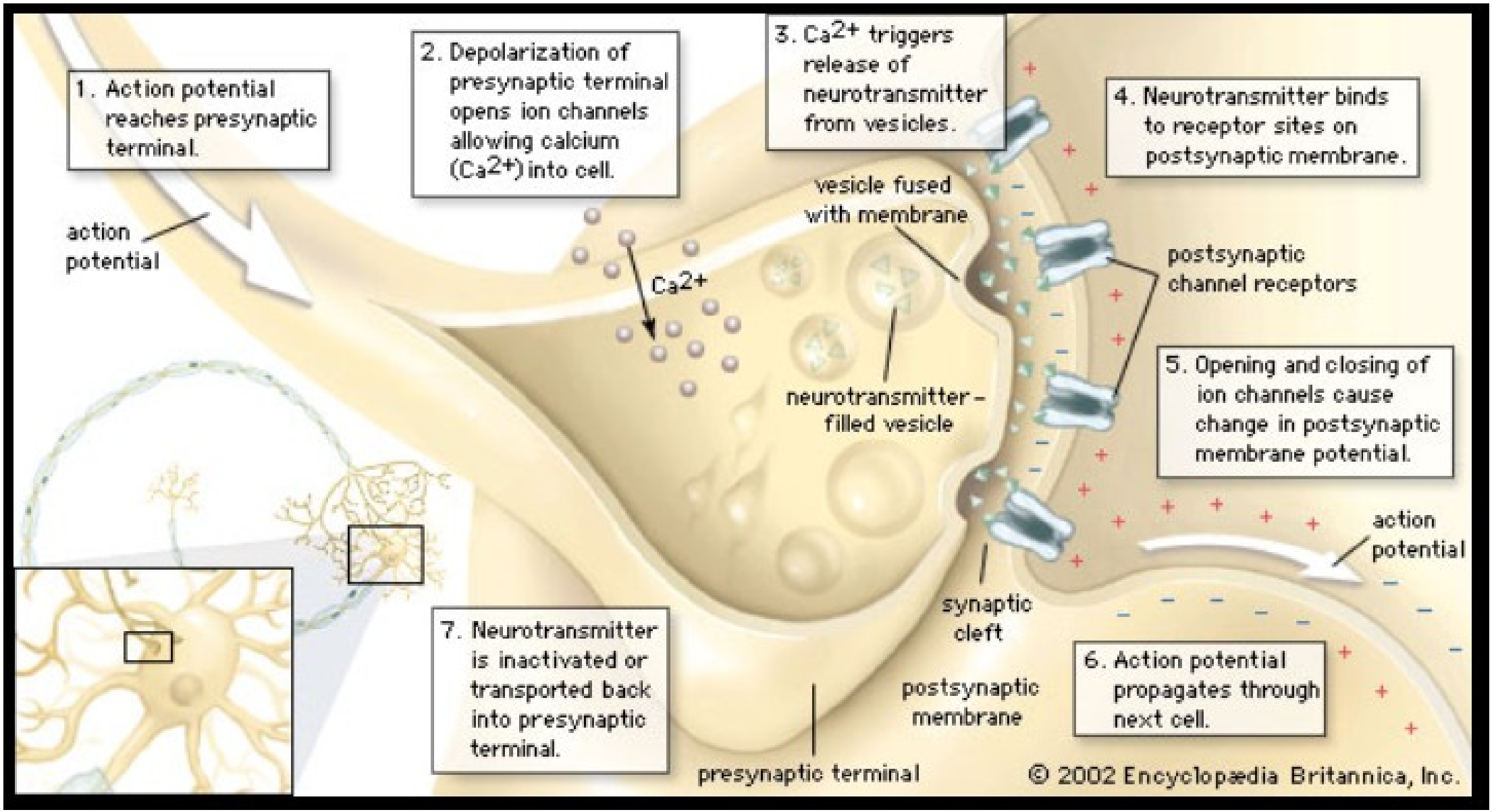


1. An Action Potential (AP) reaches the axon terminal of the presynaptic neuron.
2. Depolarisation due to the AP causes Calcium ion channels to open, allowing Ca^{2+} to enter the cell.
3. Ca^{2+} triggers vesicles containing neurotransmitter to move to the end of the terminal and release their contents.
4. Neurotransmitter crosses the synaptic cleft and binds to receptor sites on the membrane of the postsynaptic neuron dendrite.
5. Neurotransmitter/receptor complex opens Na+ channels on the dendrite, allowing Na+ to flood in, triggering depolarisation.
6. A new AP is propagated in the postsynaptic neuron.

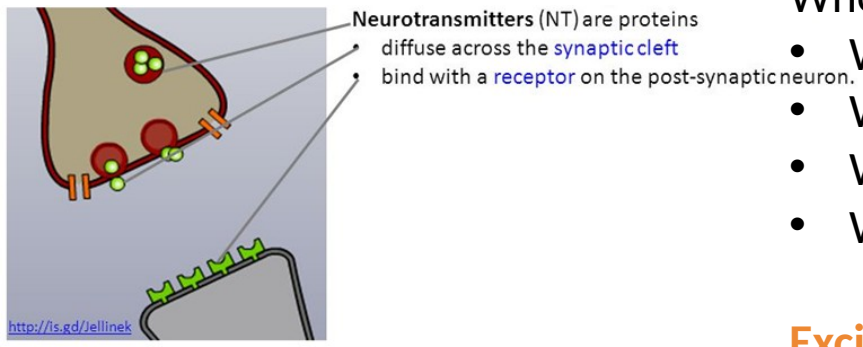
When the process is finished, the remaining neurotransmitter is reabsorbed back into the presynaptic axon terminal, for reuse.

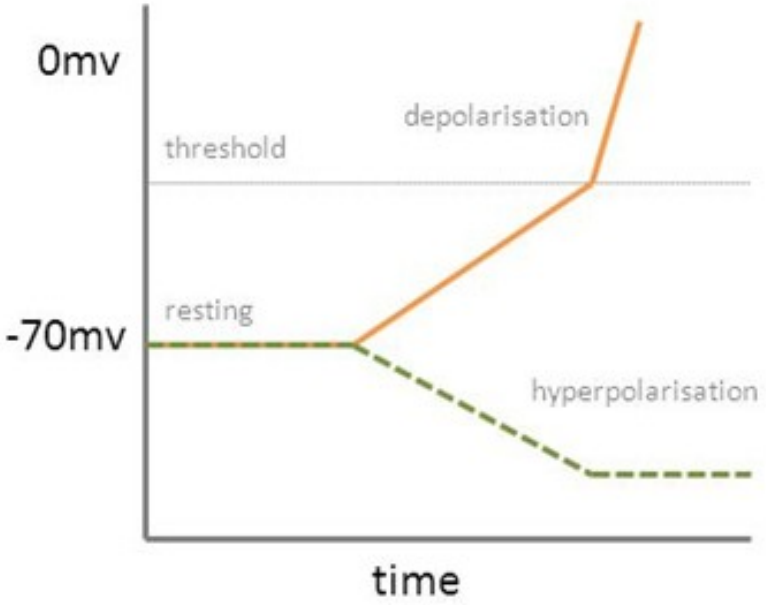
Learning Aim: Describe the sequence of events in the transmission of an action potential across a synapse.

Another diagram view...



Neurotransmitters can be excitatory or inhibitory





Whether or not the post-synaptic neuron propagates the AP depends on:

- Which NT diffuses across
- Which receptors it binds to
- Which ions flow in/out of the post-synaptic neuron
- Whether or not depolarisation reaches threshold

Excitatory NTs cause depolarisation

- NT binds, Na+ channels open, Na+ rushes in.
- Post-synaptic membrane potential depolarises
- AP propagated.

Inhibitory NTs cause hyperpolarisation

- NT binds.
- Either K+ channels open and K+ rushes out
- OR Cl- channels open, Cl- rushes in.
- Membrane potential becomes more negative – *hyperpolarisation* occurs.
- This means membrane potential is further from the threshold.
- AP is initiated more slowly, or prevented entirely.

Note: some NTs are excitatory at some synapses and inhibitory at others. More on this later – it really depends on what receptors are on the post-synaptic membrane.

Learning Aim: Describe differences in action between excitatory and inhibitory neurotransmitters.

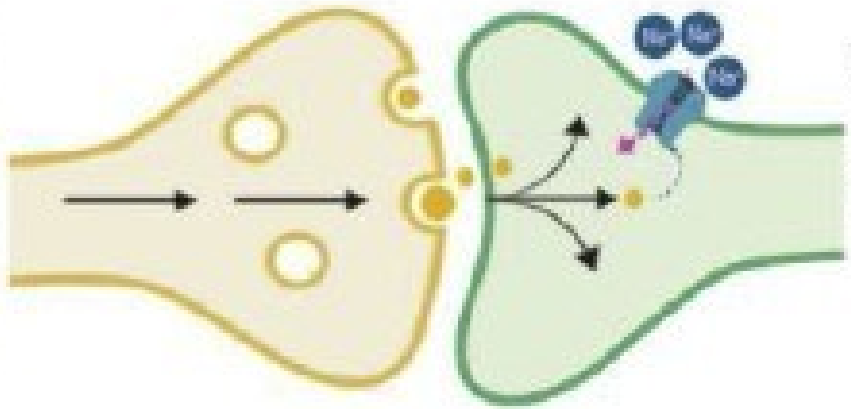
Excitatory Neurotransmitters

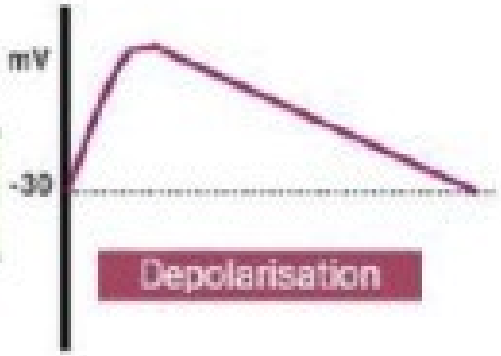
Acetylcholine

Dopamine

Norepinephrine

Glutamate





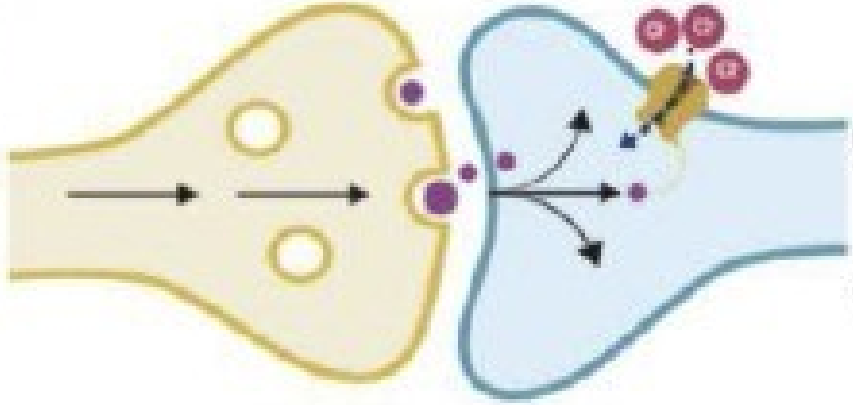
Inhibitory Neurotransmitters

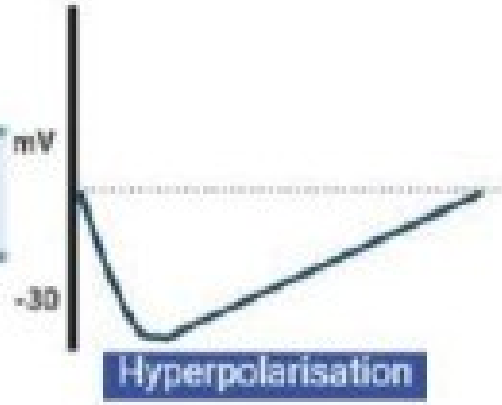
Acetylcholine

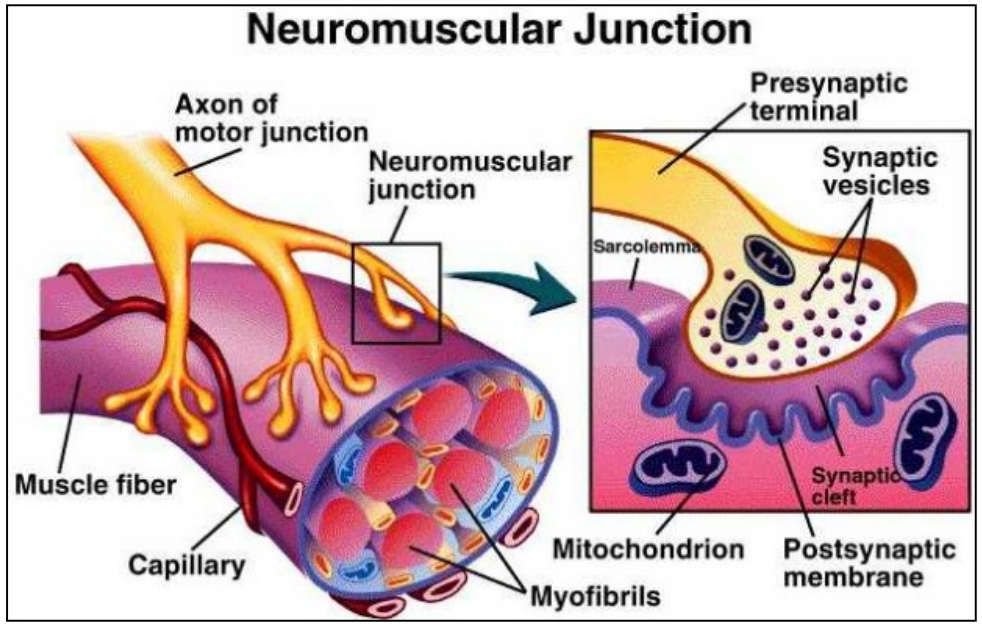
Serotonin

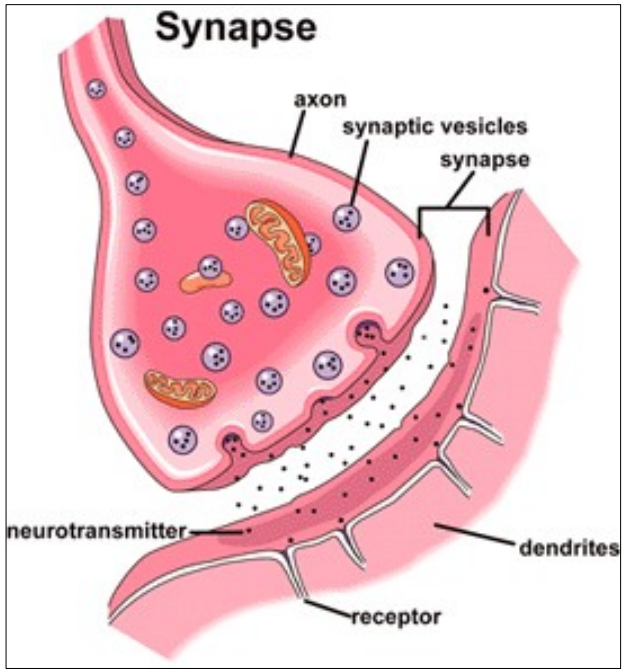
Norepinephrine

GABA









	Synapse	NMJ
Similarities	Excitable cells separated by synaptic cleft.	
	Axon terminals store and release neurotransmitter.	
	Binding of NT to receptors on the post-synaptic membrane opens ion channels, altering membrane potential.	
Differences	Junction between 2 neurons.	Junction between neuron and skeletal muscle fibre.
	Can be excitatory or inhibitory.	Only excitatory (any inhibition occurs in synapses further up the chain)

Learning Aim: Define the terms synapse, neurotransmitter and neuromuscular junction.

Other chemicals can affect the transmission at the synapse or neuromuscular junction, eg:

Stimulants

- Can stop inhibitory neurotransmitters from working, so threshold is reached more easily, having a stimulant effect.

  EG: Caffeine

- Can increase the release of an excitatory neurotransmitter from the pre-synaptic neuron, causing increased neurotransmission:

  EG: Amphetamines, Ephedrine, Pseudoephedrine, MDMA (ecstasy)

- Can prevent re-uptake of neurotransmitter from the synaptic cleft, meaning it hangs around longer, and continues to stimulate the post-synaptic neuron:

  EG: SSRIs used to treat clinical anxiety and depression

- Can increase the action of post-synaptic receptors, meaning that transmission occurs more easily.

  EG: Nicotine

Learning Aim: Discuss how stimulants, depressants and neurotoxins affect the transmission at the synapse or NMJ

Depressants

Can work in a variety of different ways. Examples include:

- Can increase the effects of an inhibitory neurotransmitter, increasing hyperpolarisation of the post-synaptic neuron

  EG: Alcohol, Valium (benzodiazepine)

- Can block or inhibit an AP by blocking ion channels in the post-synaptic neuron.

  EG: Anaesthetics

Neurotoxins

Can work in a variety of different ways. Examples include:

- Prevention of breakdown of a neurotransmitter at a synapse, causing persistent stimulation

  EG: organophosphates (snail bait and others) prevent breakdown of ACh.

- Increasing the efficiency of post-synaptic receptors

  EG: alkaloids such as strychnine.

Learning Aim: Discuss how stimulants, depressants and neurotoxins affect the transmission at the synapse or NMJ